矩阵建立、查找、取值、删除、取子式、对角矩阵、上下三角矩阵、求 逆、特征值特征向量

```
A=[1 2 3 ; 4 5 6 ; 7 8 9]
```

```
A = 3×3

     1     2     3
     4     5     6
     7     8     9
```

```
A(2,2)==A(5)
```

```
ans = logical
   1
```

```
%将第二行删除
A(2,:)=[];
disp(A)
```

```
     1     2     3
     7     8     9
```

```
A=[1 2 3 ; 4 5 6 ; 7 8 9];
A_1=A(1:2,2:3)
```

```
A_1 = 2×2

     2     3
     5     6
```

```
%求逆矩阵
syms a b c d  e f g h k;
B=[a b c;d e f ; g h k];
B_1=B^-1
```

B_1 =

$$\begin{pmatrix} \frac{f\,h-e\,k}{\sigma_1} & -\frac{c\,h-b\,k}{\sigma_1} & -\frac{b\,f-c\,e}{\sigma_1} \\ -\frac{f\,g-d\,k}{\sigma_1} & \frac{c\,g-a\,k}{\sigma_1} & \frac{a\,f-c\,d}{\sigma_1} \\ -\frac{d\,h-e\,g}{\sigma_1} & \frac{a\,h-b\,g}{\sigma_1} & -\frac{a\,e-b\,d}{\sigma_1} \end{pmatrix}$$

where

$$\sigma_1 = a\,f\,h - b\,f\,g - c\,d\,h + c\,e\,g - a\,e\,k + b\,d\,k$$

```
%求特征值，特征向量
[e,V]=eig(B)     % e:特征值，V: 特征向量
```

e =

$$\begin{pmatrix} \sigma_6 + \frac{\sigma_1\,\sigma_5}{\sigma_9} - \frac{h\,{\sigma_5}^2}{\sigma_9} & \sigma_6 - \frac{h\,{\sigma_3}^2}{\sigma_9} + \frac{\sigma_1\,\sigma_3}{\sigma_9} & \sigma_6 - \frac{h\,{\sigma_4}^2}{\sigma_9} + \frac{\sigma_1\,\sigma_4}{\sigma_9} \\ \frac{g\,{\sigma_5}^2}{\sigma_9} - \frac{\sigma_2\,\sigma_5}{\sigma_9} - \sigma_7 & -\sigma_7 + \frac{g\,{\sigma_3}^2}{\sigma_9} - \frac{\sigma_2\,\sigma_3}{\sigma_9} & -\sigma_7 + \frac{g\,{\sigma_4}^2}{\sigma_9} - \frac{\sigma_2\,\sigma_4}{\sigma_9} \\ 1 & 1 & 1 \end{pmatrix}$$

where

$$\sigma_1 = b\,g + e\,h + h\,k$$

$$\sigma_2 = a\,g + d\,h + g\,k$$

V =

$$\begin{pmatrix} \sigma_3 & 0 & 0 \\ 0 & \frac{a}{3}+\frac{e}{3}+\frac{k}{3}-\frac{\sigma_2}{2}-\frac{\sigma_3}{2\,\sigma_2}-\sigma_1 & 0 \\ 0 & 0 & \frac{a}{3}+\frac{e}{3}+\frac{k}{3}-\frac{\sigma_2}{2}-\frac{\sigma_3}{2\,\sigma_2}+\sigma_1 \end{pmatrix}$$

$$\sigma_3 = \frac{a}{3}+\frac{e}{3}+\frac{k}{3}+\frac{\sigma_{10}}{2}-\frac{\sigma_{11}}{2\,\sigma_{10}}-\sigma_8$$

$$\sigma_4 = \frac{a}{3}+\frac{e}{3}+\frac{k}{3}-\frac{\sigma_{10}}{2}-\frac{\sigma_{11}}{2\,\sigma_{10}}+\sigma_8$$

where

$$\sigma_5 = \frac{a}{3}+\frac{e}{3}+\frac{k}{3}+\sigma_{10}+\frac{\sigma_{11}}{\sigma_{10}}$$

$$\sigma_6 = \frac{f\,h^2 + c\,g\,h - b\,g\,k - e\,h\,k}{}$$

$$\sigma_1 = \frac{\sqrt{3}\left(\sigma_2 - \frac{\sigma_3}{\sigma_2}\right) i}{2}$$

掌握if、switch 选择语句的基本用法

```
score=70;
if score >= 90
    G='A';
    fprintf('Excelent!');
else
    G='B';
    fprintf('Not Very Bad.');
end
```

```
Not Very Bad.
```

```
switch G
    case 'A'
        fprintf('Excelent');
    case 'B'
        fprintf('Not Very Bad.');
end
```

```
Not Very Bad.
```

掌握for、while 循环语句的基本用法

```
%以求阶乘为例
n=1;
sum=1;
if n == 0
    sum=1;
else
    for c = 1:1:n
        sum=sum*c;
    end
end
disp(sum);
```

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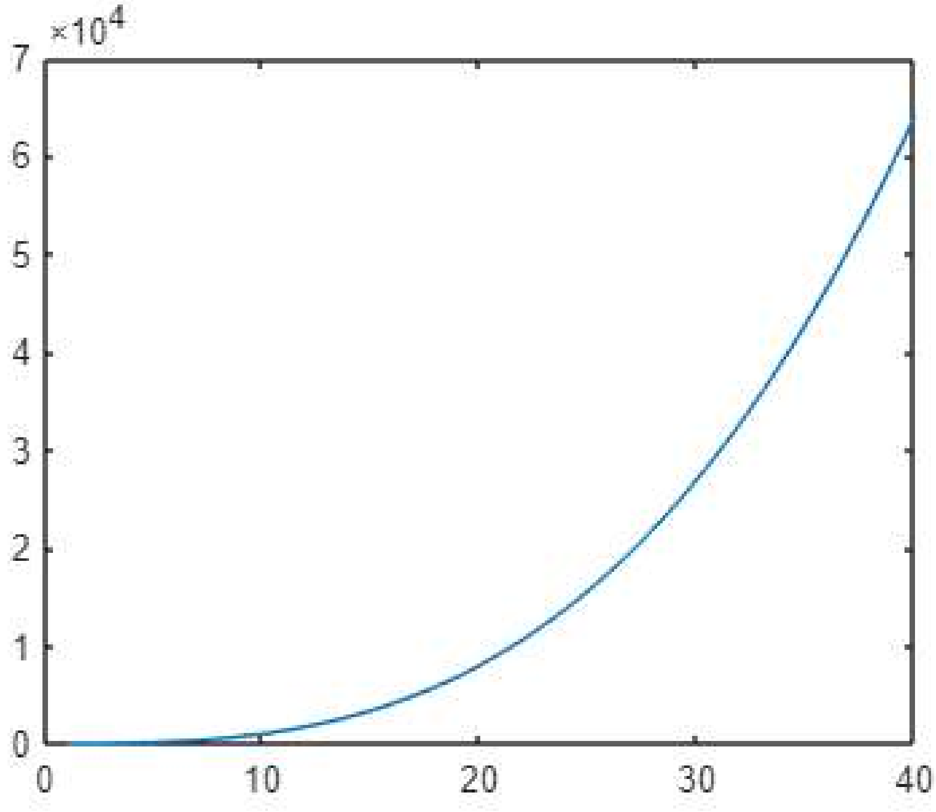
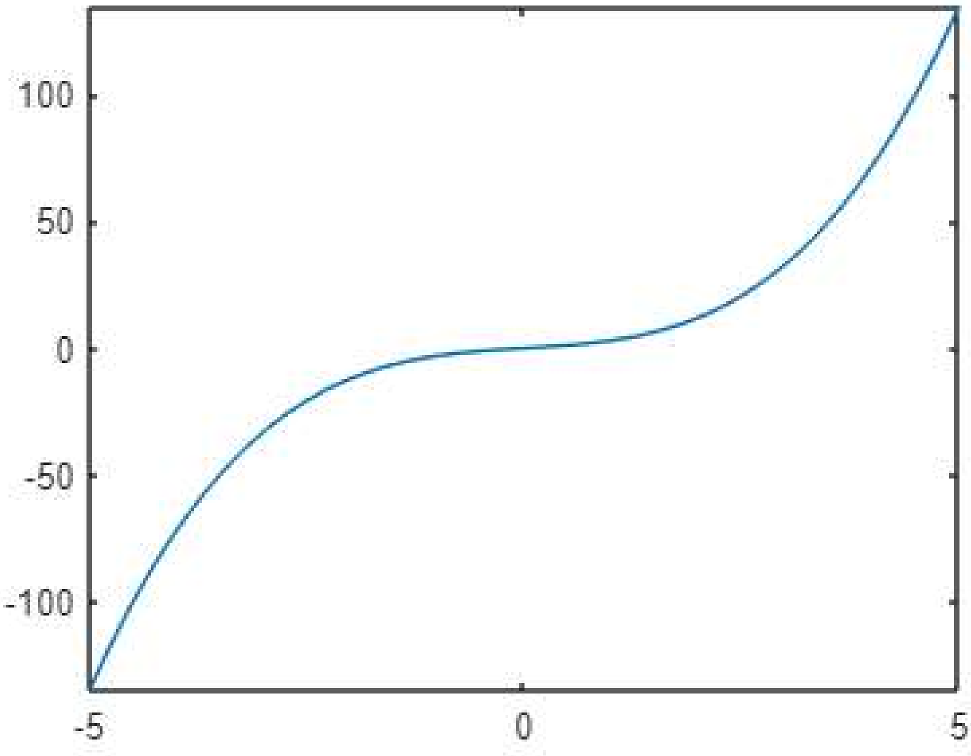
使用plot、fplot，fimplicit 绘制二维图，会使用surf、mesh，fsurf，fimplicit3 等绘制曲面图，并能设置图像的标题、图例以及坐标轴的标记等操作。

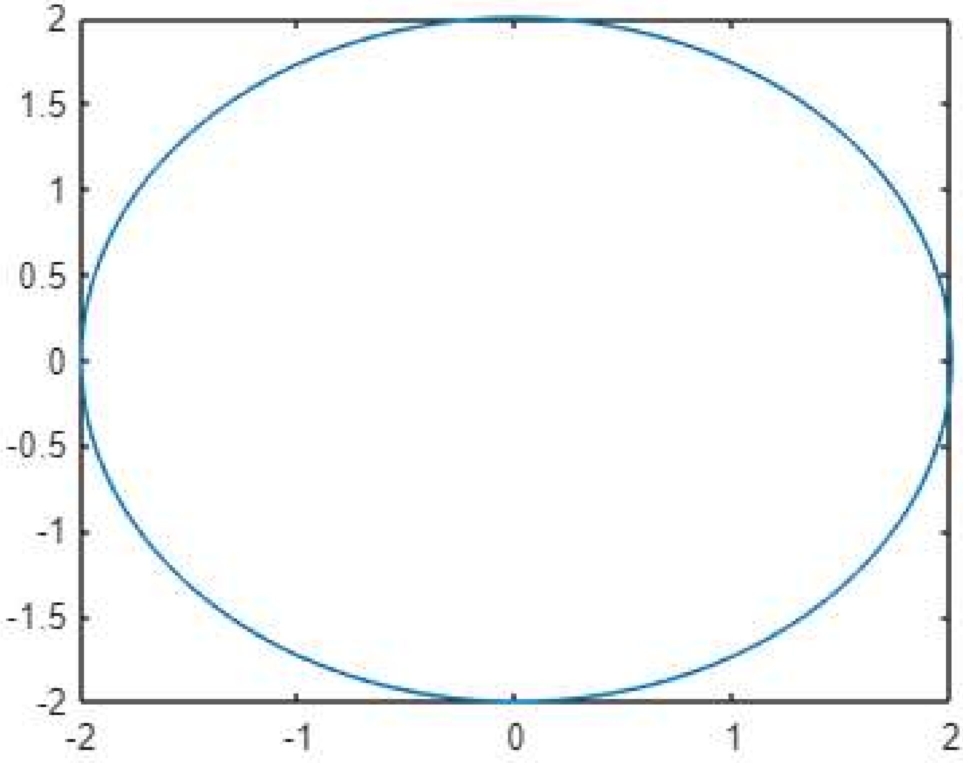
```
x=1:0.02:40;
y=x.^3+2*x;
plot(x,y)
```



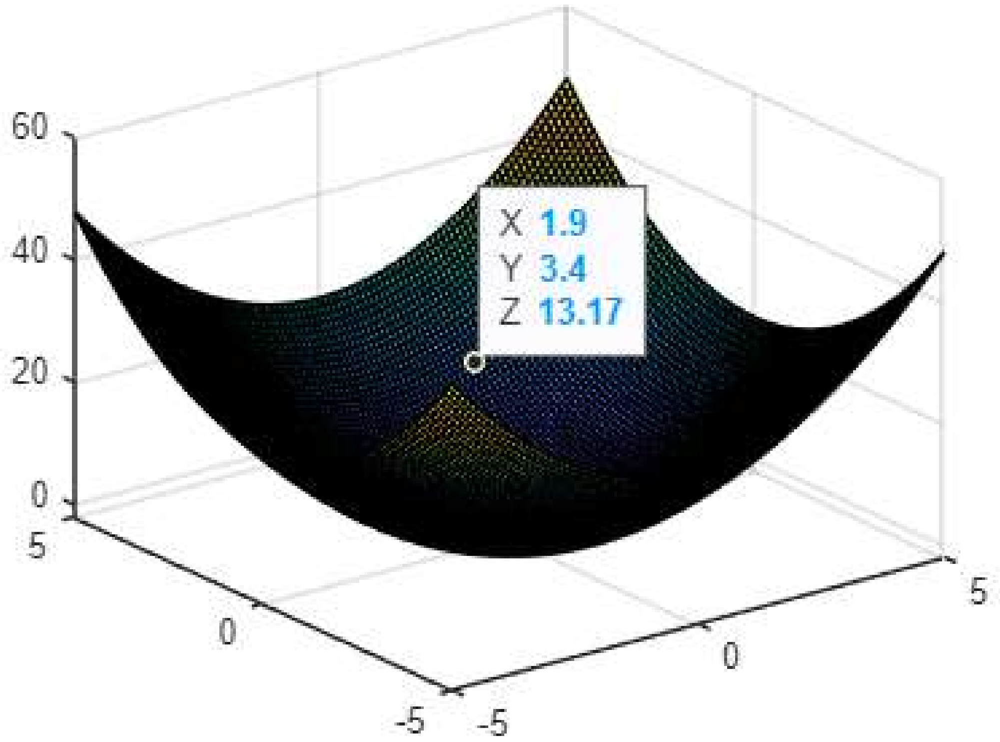

```
f=@(x)x.^3+2*x;
fplot(f)
```



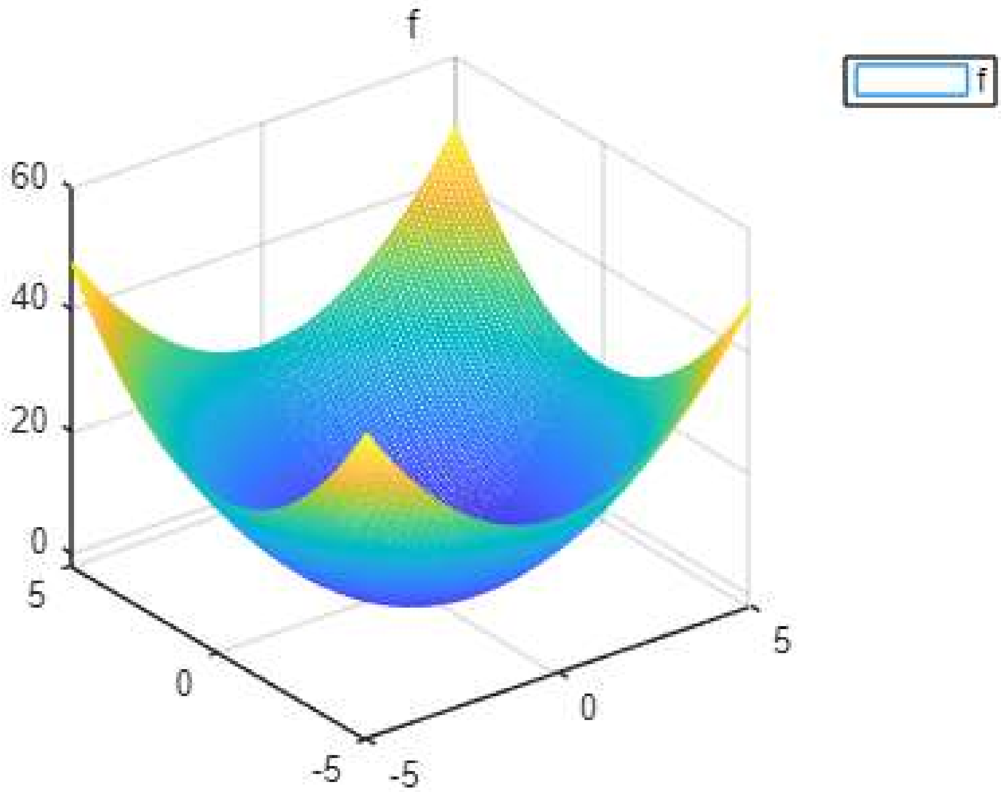

```
f=@(x,y)x.^2+y.^2-4;
fimplicit(f)
```




```
x=-5:0.1:5;
y=-5:0.1:5;
[X,Y]=meshgrid(x,y);
Z=X.^2+Y.^2-2;
surf(X,Y,Z)
```




```
x=-5:0.1:5;
y=-5:0.1:5;
[X,Y]=meshgrid(x,y);
Z=X.^2+Y.^2-2;
mesh(X,Y,Z)
title('f')
legend('f')
```



使用sum、max、min、sort、std、cov、corrcoef、eig，det

```
clear;clc;
A=[1 2 3;4 5 6;7 8 9];
sum(A)
```

```
ans = 1×3
    12    15    18
```

```
max(A)
```

```
ans = 1×3
     7     8     9
```

```
B=[2 4 6 ;7 9 3; 6 8 1];
sort(B)
```

```
ans = 3×3
     2     4     1
     6     8     3
     7     9     6
```

```
std(A)
```

```
ans = 1×3
     3     3     3
```

计算一个正态分布的随机矩阵的相关系数和 p 值，其中添加的第四列等于其他三列之和。由于 A 的最后一列是其他列的线性组合，第四个变量与其他三个变量中的每一个之间建立了相关性。因此，P 的第四行和第四列包含非常小的 p 值，将其标识为显著相关。

```
A = randn(50,3);
A(:,4) = sum(A,2);
[R,P] = corrcoef(A)
```

```
R = 4×4
    1.0000    0.1135    0.0879    0.7314
    0.1135    1.0000   -0.1451    0.5082
    0.0879   -0.1451    1.0000    0.5199
    0.7314    0.5082    0.5199    1.0000
P = 4×4
    1.0000    0.4325    0.5438    0.0000
    0.4325    1.0000    0.3146    0.0002
    0.5438    0.3146    1.0000    0.0001
    0.0000    0.0002    0.0001    1.0000
```

```
f=[3 -5 -7 5 6];
g=[0 0 3 5 -3];
f+g
```

```
ans = 1×5
     3    -5    -4    10     3
```

```
%乘法用conv计算
conv(f,g)
```

```
ans = 1×9
     0     0     9     0   -55    -5    64    15   -18
```

```
h=[2,1,3];
[Q,r]=deconv(f,h) %Q是商，r是余数
```

```
Q = 1×3
    1.5000   -3.2500   -4.1250
r = 1×5
         0         0         0   18.8750   18.3750
```

```
roots(h)    % 求根
```

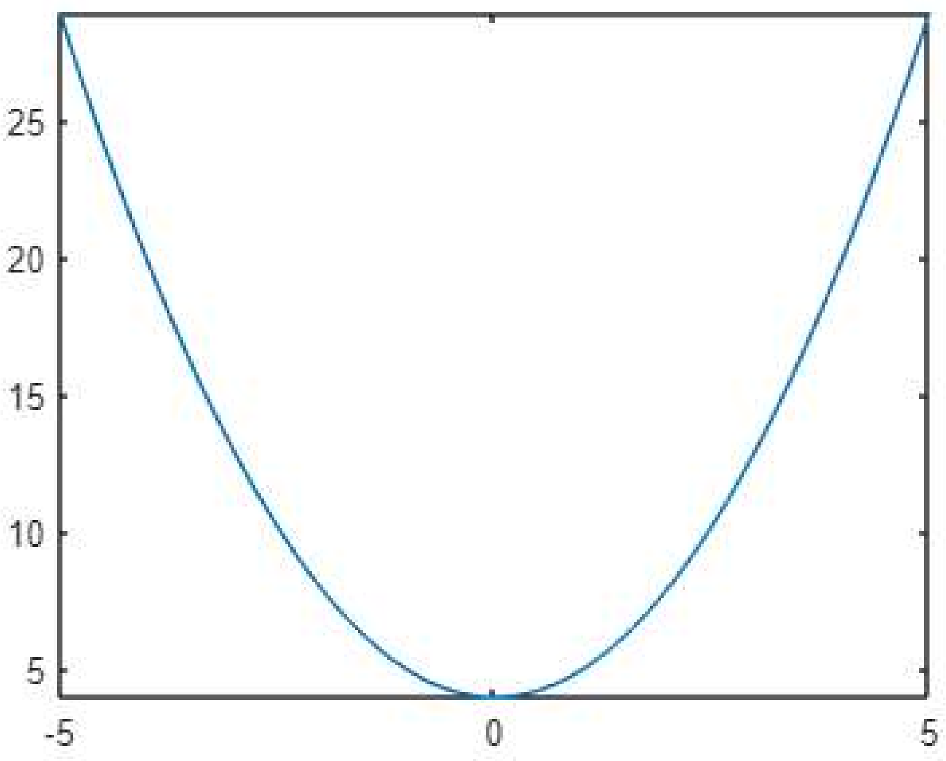
```
ans = 2×1 complex
  -0.2500 + 1.1990i
  -0.2500 - 1.1990i
```

```
% 矩阵多项式
A=[2 0 0;1 1 1;1 -1 3];
f=[1 -2 0 1 -1];
y=polyvalm(f,A)
```

```
y = 3×3
     1     0     0
     9    -8     9
     9    -9    10
```

用integral、integral2、integral3 求积分积分的数值解，求重积分时，当上 下限是函数时，需懂得上下限函数的表示方法

```
clear;clc;
f=@(x)x.^2+4;
fplot(f)
```




```
q=integral(f,0,1)
```

```
q = 4.3333
```

```
%二重积分
fun = @(x,y) 1./( sqrt(x + y) .* (1 + x + y).^2 );
q=integral2(fun,0,1,0,1)
```

```
q = 0.3695
```

用rref化简矩阵

```
A=magic(3);
q=rref(A)
```

```
q = 3×3
     1     0     0
     0     1     0
     0     0     1
```